# Mechanical Engineering Project Portfolio

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# Cycloidal Drive | June 2026 – July 2026

**Related Skills:** SolidWorks, Siemens NX, Experimentation

**Purpose:** Design, build and test a cycloidal drive which will be used to increase the transmission ratio of a Nema 17 stepper motor rated for a maximum torque of 59 N*cm. The goal transmission ratio of this cycloidal drive was 15:1.

**Solution: Design**

- Calculated the necessary geometry to achieve a 15:1 transmission ratio, including the necessary curve equation for the cycloidal disk, roller count, cycloidal lobe count, and other critical dimensions of the various components.
- Modeled a complete assembly of the drive in SolidWorks. Two cycloidal disks used to ensure eccentric stability. 5mm dowel pins were selected for the outer rollers because of their tight tolerances, various sized ball bearings were used to properly transmit the rotational motion and maintain alignment between the shafts, and aluminum spacers were used for the output rollers to allow for screw connection between output shaft and support flange. M3 screws were the only fasteners chosen in the design.
- Ordered and 3D printed all the parts and maintained proper BOM. The cycloidal disks were printed out of ABS filament due to its increased strength and impact resistance, everything else was printed out of PLA. 4 wall layers were used for every part, the disks were made using 100% infill, while everything else was 35% infill.
- Assembled various prototypes and found some issues which required fixing. Initially the output shaft did not have the proper eccentricity because it wasn't properly aligned, and the disks would wobble when spun causing lockup, so the shaft had to be redesigned to ensure proper eccentricity and alignment. These prototypes helped iterate and verify part dimensions to ensure proper tolerances were kept and documented. Revisions were made until proper functionality was achieved.

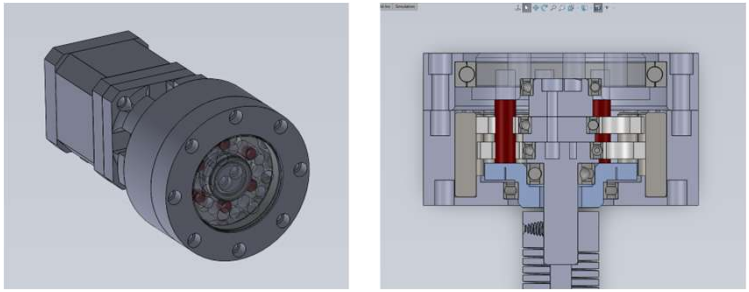
For Transmission Ratio of i = 15

$$i = 15 = \frac{n}{N - n}$$

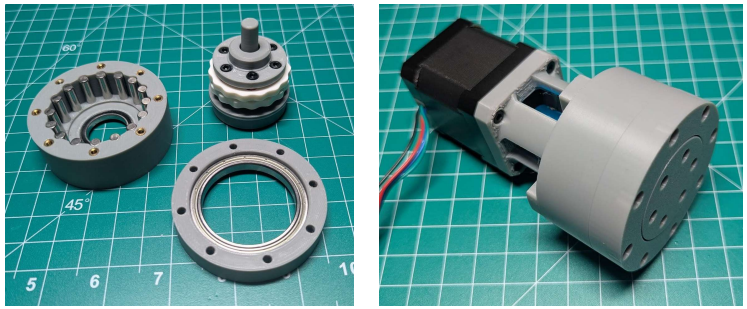
Chosen Roller Count: N = 16

Lobe Count $n = N - 1 = 15$

Calculated Eccentricity e = 0.563



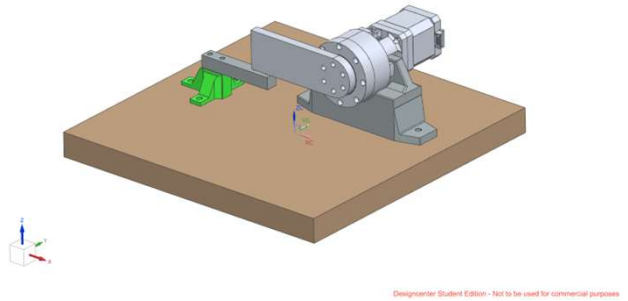
Assembly model in SolidWorks



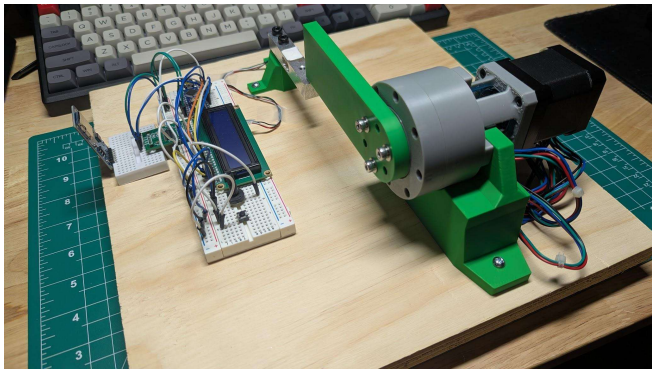
Physical Assembly

**Solution Continued: Testing**

- Modeled a moment-arm test jig in Siemens NX to measure the cycloidal drive's output force via a load cell. Since the jig and drive stand were 3D printed, stiffness was a key design driver to preserve reading accuracy. A 100mm moment arm length was chosen for calculation simplicity, and the load cell was leveled to ensure the arm applied a near-vertical force.
- 3D printed all jig components at 50% infill with 4 wall loops for added strength, then wired and assembled the test circuit: an Arduino Nano as the primary microcontroller, a 16x2 LCD, a 1KΩ potentiometer, a tare button, a 20kg load cell, and an HX711 module to amplify and digitize the load cell's output. A custom DRV8825 2A breakout board powered the NEMA 17 stepper motor.
- Calibrated the load cell against a known weight, then slowly actuated the stepper motor via serial commands to repeatedly load and unload the moment arm against the load cell until slip occurred. The LCD tracked and displayed the peak force reading for each run, and was reset between trials, to capture the drive's maximum holding force.



Testing Jig Model in Siemen's NX



Final Testing Jig Assembly

**Results:**

Eight trials were conducted with the NEMA 17 stepper at its 2A current rating, each producing a maximum holding force reading from the load cell. Averaging these eight values yielded a mean maximum holding force of 15.642N. Multiplying by the 100mm moment arm length (converted to 10cm) gives an experimental maximum holding torque of 156.421N·cm. Compared to the theoretical 885N·cm output expected from the 15:1 drive ratio, this corresponds to a total drive efficiency of 17.67%.

| Trial | Force (N) |
|---|---|
| 1 | 15.587 |
| 2 | 15.991 |
| 3 | 15.889 |
| 4 | 15.685 |
| 5 | 15.759 |
| 6 | 15.125 |
| 7 | 15.545 |
| 8 | 15.556 |

**Future Work :**

The cycloidal drive fell short of its 15:1 transmission goal, achieving only 17.67% of the theoretical output, an 83% shortfall driven by several compounding factors. On the design side, switching the rollers from dowel pins to 5mm ball bearings, improving lubrication, and adjusting the layout to prevent lockup from over-tightened screws could reduce friction and unnecessary energy loss. On the testing side, measurement error from both the load cell and the Arduino Nano may have contributed, along with possible misalignment of the moment arm relative to the load cell's center axis. Future test should characterize dynamic torque conditions.

# Senior Design Project: Hybrid Torque-Vectoring Vehicle | August 2025 – May 2026

**Related Skills**: SolidWorks, Mechanical/Electrical Integration, DFA, Data Acquisition, Hand Tools

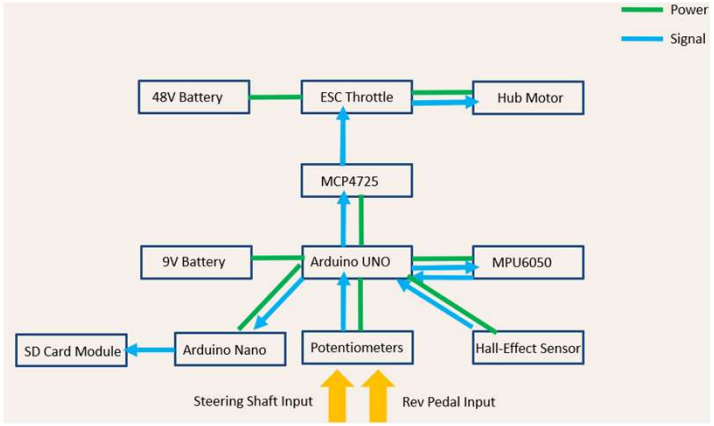
**Purpose:** The purpose of my Senior Design Project was to design and manufacture a hybrid vehicle capable of torque-vectoring and reaching 25 miles per hour.

**Solution: Design**

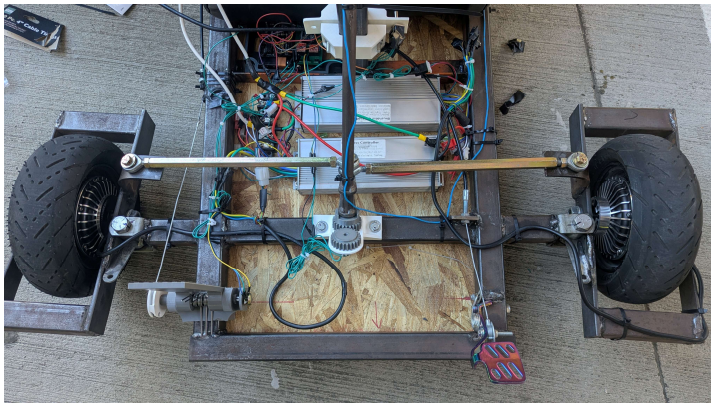
- Worked in team of 5 as electrical lead on designing the electrical systems for a split-drive hybrid vehicle , configuring dual front 48V 1000W hub motors and a rear internal combustion engine to enable active front-wheel torque vectoring at a 25-mph target velocity.
- Selected 10-inch wheels for sharper turns, requiring 840.3 RPM to meet the 25mph goal — met by the chosen 48V 1000W hub motors, rated for 1042 RPM. Calculated a ~50A max current draw from the motors' power/voltage rating at 85% efficiency and selected a 48V battery with a 60A continuous BMS for a 1.2 safety margin.
- Built an Arduino Uno R4 control loop that polls the MPU6050 gyro/accelerometer and B10K potentiometers over $I^2C$ to determine steering angle and throttle, then outputs the corresponding signals through dual MCP4725 DACs to drive the 48V 1000W ESC motor controllers. Sensor data was recorded onto a 32GB microSD card with an SD card module.
- While the R4 was initially chosen as the sole microcontroller, an Arduino Nano was later added to resolve a freezing issue the R4 experienced when repeatedly opening and closing the log file. The Nano kept the file open continuously and logged the readings sent by the R4.
- Worked with group in assembling the vehicle, including fastening, stick-welding, and grinding of chassis components. Furthermore, wired and assembled all electrical components including sensors, hub motors and battery, ensuring safe and secure connections were established.

Motor Current Draw (single)

$$I_{motor} = \frac{P}{V \times \eta} = \frac{1000W}{48V \times 0.85} = 24.5A$$




Torque Vectoring Control System



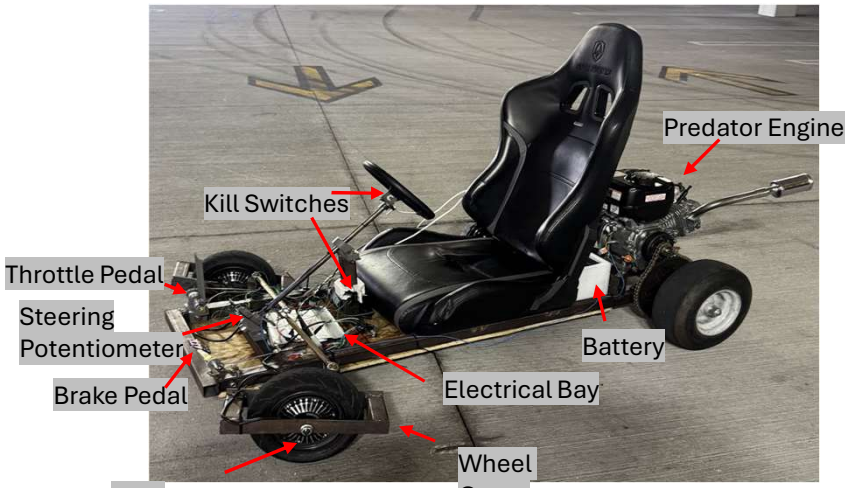
Final Wiring

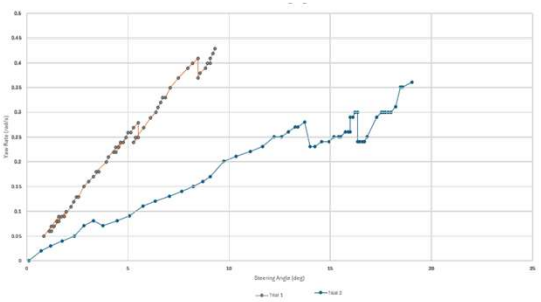
**Solution: Testing**

- Calibrated the B10K potentiometers to ensure throttle and steering angle voltages were accurately mapped to their physical inputs and calibrated the Hall effect sensor to ensure accurate rear-axle speed readings.
- For testing, two trials were conducted with a designated driver: the first in hybrid (gas and electric) configuration, and the second in gas-only configuration. These test were done on the same designated track.

**Results:** Both trials measured the same turn on the track, isolating the effect of torque vectoring on yaw rate and angular acceleration relative to steering angle. At a steering angle of 8°, the hybrid (torque-vectoring) configuration produced a yaw rate of 0.43 rad/s, more than double the 0.17 rad/s measured at the same angle in gas-only configuration. Angular acceleration followed a similar pattern: the hybrid configuration peaked at 1.62 rad/s$^2$ near 8°, a value the gas-only configuration didn't reach until roughly 17°. Torque vectoring therefore produced a higher yaw response at a lower steering angle, confirming its intended effect.

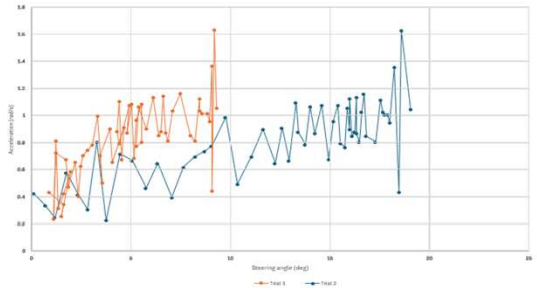
**Future Work:** Future iterations of the vehicle will focus on integrating a mechanical suspension system and vibration suppression to improve driver control and eliminate the abrupt noise spikes seen in the MPU6050 accelerometer data and physical misalignments in the potentiometer mounts. Additionally, to resolve the hardware data-logging bottleneck, the firmware can be upgraded to a proper dual-core microcontroller architecture to efficiently balance high-frequency $I^2C$ sensor polling with data write cycles and reduce the need for two one-core microcontrollers.



Yaw Rate (rad/s)
vs
Steering angle (deg)

Angular Acceleration (rad/$s^2$)
vs
Steering angle (deg)



# Multi-Punch Pneumatic Fixture | June 2025 – August 2025

**Related Skills**: Autodesk Inventor, Technical Drawings, Milling Machine, Project Planning, Communication

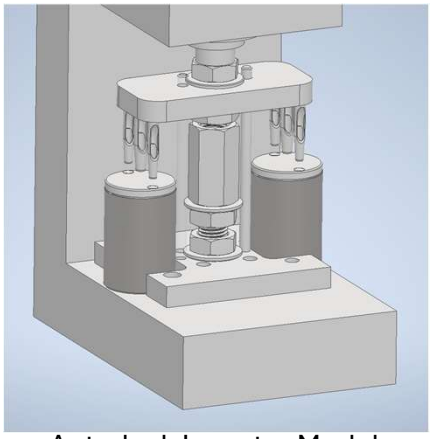
**Purpose:** An optimized manufacturing method was required to eliminate manual steps and improve efficiency when making cutouts on plastic sleeves. The objective was to engineer a hardware solution capable of significantly increasing production output while repeatedly maintaining manufacturing tolerances within ±0.002 inches**.**

**Solution:**

- Developed initial design concepts in Autodesk Inventor to establish geometric layouts, selecting a punch attachment which could be mounted on an existing shop pneumatic press.
- Generated detailed technical drawings to guide the component fabrication process during the machining process.
- Fabricated the base alignment plate out of a PVC plate and punch holder from 1040 steel under machinist supervision using a manual milling machine, drill press, and angle grinder.
- Assembled the final fixture and performed tolerance checks to verify precise part fitment and smooth mechanical functionality.
- Conducted systematic testing on the completed assembly to evaluate operational accuracy and process efficiency, making accurate cutouts of 100 parts.

**Results:** Following validation testing, production throughput successfully increased from 1 part every 48 seconds to 2 parts every 8 seconds. This optimization represents an 8x increase in total production output while flawlessly maintaining the required geometric tolerances.

**Future Work:** Future iterations will focus on scaling the tooling architecture to incorporate another 3-punch set, expanding total cycle part capacity. Additionally, implementing tighter geometric dimensioning and tolerancing (GD&T) constraints during fabrication will further minimize wobble and optimize punch alignment.



Autodesk Inventor Model



Final Completed Fixture Assembly